

Task: 6: Automated Logging using Triggers & Views

Aim: To implement an automated logging system using database triggers that records every INSERT and UPDATE operation, and to generate daily activity reports using database views.

Algorithm:

1. Start the program
2. Insert or update employee data
3. Trigger activates automatically on INSERT/UPDATE
4. Store log details in logs table
5. Fetch log data from database
6. Use view to generate daily report
7. Display logs and report on screen
8. Stop the program

Program:

SQL:

```
CREATE DATABASE IF NOT EXISTS audit_db;

USE audit_db;

-- Employees table

CREATE TABLE employees (

    id INT AUTO_INCREMENT PRIMARY KEY,

    name VARCHAR(50),

    salary INT

);

-- Logs table

CREATE TABLE logs (

    log_id INT AUTO_INCREMENT PRIMARY KEY,

    action VARCHAR(50),

    emp_id INT,

    emp_name VARCHAR(50),

    time TIMESTAMP DEFAULT CURRENT_TIMESTAMP

);

-- Trigger: INSERT

CREATE TRIGGER after_insert_emp
```

```

AFTER INSERT ON employees
FOR EACH ROW
INSERT INTO logs(action, emp_id, emp_name)
VALUES ('INSERT', NEW.id, NEW.name);
-- Trigger: UPDATE
CREATE TRIGGER after_update_emp
AFTER UPDATE ON employees
FOR EACH ROW
INSERT INTO logs(action, emp_id, emp_name)
VALUES ('UPDATE', NEW.id, NEW.name);
-- VIEW (IMPORTANT — your previous error was here)
CREATE VIEW daily_report AS
SELECT DATE(time) AS date, action, emp_name
FROM logs
ORDER BY time DESC;

```

db.php:

```

<?php
$conn = new mysqli("localhost", "root", "", "audit_db");
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>

```

activity.php:

```

<?php
include 'db.php';
// INSERT
if (isset($_POST['add'])) {

```

```

$name = $_POST['name'];
$salary = $_POST['salary'];
$conn->query("INSERT INTO employees(name, salary) VALUES('$name', '$salary')");
}
// UPDATE
if (isset($_POST['update'])) {
    $id = $_POST['id'];
    $salary = $_POST['salary'];
    $conn->query("UPDATE employees SET salary='$salary' WHERE id='$id'");
}
// Fetch logs
$logs = $conn->query("SELECT * FROM logs");
// Fetch view
$report = $conn->query("SELECT * FROM daily_report");
?>
<!DOCTYPE html>
<html>
<head>
    <title>Audit Logging System</title>
    <style>
        body { font-family: Arial; background: #f2f2f2; }
        .box { width: 80%; margin: auto; background: white; padding: 20px; }
        input, button { padding: 8px; margin: 5px; }
        table { width: 100%; border-collapse: collapse; margin-top: 15px; }
        th, td { border: 1px solid black; padding: 8px; text-align: center; }
        h2 { color: blue; }
    </style>
</head>
<body>

```

```

<div class="box">
<h2>Add Employee</h2>
<form method="POST">
    <input type="text" name="name" placeholder="Name" required>
    <input type="number" name="salary" placeholder="Salary" required>
    <button name="add">Add</button>
</form>
<h2>Update Employee</h2>
<form method="POST">
    <input type="number" name="id" placeholder="ID" required>
    <input type="number" name="salary" placeholder="New Salary" required>
    <button name="update">Update</button>
</form>
<h2>Logs (Trigger Output)</h2>
<table>
<tr><th>ID</th><th>Action</th><th>Employee</th><th>Time</th></tr>
<?php while($row = $logs->fetch_assoc()) { ?>
<tr>
<td><?php echo $row['log_id']; ?></td>
<td><?php echo $row['action']; ?></td>
<td><?php echo $row['emp_name']; ?></td>
<td><?php echo $row['time']; ?></td>
</tr>
<?php } ?>
</table>
<h2>Daily Report (View)</h2>
<table>
<tr><th>Date</th><th>Action</th><th>Employee</th></tr>
<?php while($row = $report->fetch_assoc()) { ?>

```

```

<tr>

<td><?php echo $row['date']; ?></td>

<td><?php echo $row['action']; ?></td>

<td><?php echo $row['emp_name']; ?></td>

</tr>

<?php } ?>

</table>

</div>

</body>

</html>

```

OUTPUT:

The screenshot shows a web browser window with the URL `localhost / 127.0.0.1 / audit_db` and the page title `Audit Logging System`. The browser address bar shows `localhost/student_project/activity.php`. The page content is as follows:

Add Employee

Form fields:

Update Employee Salary

Form fields:

Logs (Trigger Output)

ID	Action	Employee	Time
1	INSERT	John	2026-04-18 23:12:51

Daily Report (View)

Date	Action	Employee
2026-04-18	INSERT	John

Result:

Thus, the Automated Logging using Triggers & Views is done successfully and output generated .